

1 Basic Numbers & Arithmetic

1.1 1. The Number 0

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x - \sum_{k=0}^{\infty} \frac{1}{k!}$$

- **Translation:** 0
- **The De-obfuscation:** Both terms are just different definitions of Euler's number (e). Subtracting $e - e$ gives 0.

1.2 2. The Number 1

$$\prod_{p \text{ prime}} \left(1 - \frac{1}{p^2}\right)^{-1} \cdot \frac{6}{\pi^2}$$

- **Translation:** 1
- **The De-obfuscation:** The infinite product is Euler's solution to the Basel Problem ($\frac{\pi^2}{6}$). Multiplying $\frac{\pi^2}{6} \cdot \frac{6}{\pi^2}$ equals 1.

1.3 3. The Number 1 (Euler's Identity variant)

$$- \exp(i\pi)$$

- **Translation:** 1
- **The De-obfuscation:** Euler's formula states $e^{i\pi} = -1$. Negating it gives $-(-1) = 1$.

1.4 4. The Number 2 (Set Theory / Peano)

$$|\{\emptyset, \{\emptyset\}\}|$$

- **Translation:** 2
- **The De-obfuscation:** In set theory (Von Neumann ordinals), numbers are defined by sets containing sets. This is a set containing two elements: the empty set, and a set with the empty set inside it. The cardinality $|\cdot|$ counts them: 2.

1.5 5. Simple Addition ($x + y$)

$$(x \oplus y) + 2(x \wedge y)$$

- **Translation:** $x + y$
 - **The De-obfuscation:** This comes from Mixed Boolean Arithmetic (MBA). It states that $x + y$ equals the XOR sum (bits where they differ) plus twice the AND sum (the carry bits).
-

2 Basic Operations & Functions

2.1 6. Multiplying a Number by 2

$$\int_0^x 2 dt$$

- **Translation:** $2x$
- **The De-obfuscation:** Integrating the constant 2 from 0 to x is just $2 \cdot x$.

2.2 7. Taking the Average of Two Numbers (a and b)

$$\arg \min_x ((x - a)^2 + (x - b)^2)$$

- **Translation:** $\frac{a+b}{2}$ (The mid-point / average)
- **The De-obfuscation:** Find the point x that minimizes the squared distance to a and b . That point is always the exact middle.

2.3 8. A Constant Function ($f(x) = 1$)

$$\frac{d}{dx} (x + \sin^2(x) + \cos^2(x)) - 1$$

- **Translation:** 0
- **The De-obfuscation:** $\sin^2(x) + \cos^2(x) = 1$. So the expression is $\frac{d}{dx}(x + 1) - 1 = 1 - 1 = 0$.

2.4 9. Absolute Value ($|x|$)

$$\sqrt{x^2}$$

- **Translation:** $|x|$ (Distance from zero)
 - **The De-obfuscation:** Squaring any number makes it positive; taking the principal square root keeps it positive.
-

3 Probability & Data (Neuro-Adjacent)

3.1 10. Counting How Many Items Match a Condition

$$\sum_{i=1}^N \mathbb{I}(x_i > 0) \quad \text{where } \mathbb{I}(A) = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{otherwise} \end{cases}$$

- **Translation:** *"Count how many positive numbers are in the list."*
- **The De-obfuscation:** $\mathbb{I}$ is an "indicator function"—an if statement that returns 1 or 0. The summation $\sum$ acts as a counter loop.

3.2 11. Normalizing Data into Percentages (Softmax)

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

- **Translation:** *"Turn a set of scores into probabilities that add up to 100%."*
- **The De-obfuscation:** Exponentiating e^{z_i} ensures all values are positive, and dividing by the sum forces the total to equal 1.

3.3 12. Normal Distribution / Bell Curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- **Translation:** *"Scores cluster symmetrically around the average (μ), decaying smoothly as you move further away."*
 - **The De-obfuscation:** The fraction at the front is just a constant multiplier so the total area equals 1. The core mechanism is $e^{-(\text{distance})^2}$.
-

4 Matrices & Logic

4.1 13. Swapping Two Variables (a and b)

$$\begin{bmatrix} a \\ b \end{bmatrix} \leftarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

- **Translation:** `temp = a; a = b; b = temp;`
- **The De-obfuscation:** Multiplying a 2D vector by a anti-diagonal permutation matrix flips the top and bottom values.

4.2 14. Checking if x is Even or Odd

$$x \pmod{2} \equiv 0$$

- **Translation:** *"Is x an even number?"*
- **The De-obfuscation:** The remainder when x is divided by 2 is 0.

4.3 15. The Identity Matrix Operations

$$\det(I_n) = 1^n$$

- **Translation:** `1 = 1`
- **The De-obfuscation:** The determinant of an $n \times n$ identity matrix (a grid with 1s down the diagonal) is simply 1.